Subject Description Form

Subject Code	**SFT420DD**
Subject Title	**Advanced Digital Design and Visualisation**
Credit Value	**3**
Level	**4**
Pre-requisite/ Co-requisite/ Exclusion	SFT302DD – 3D CAD in Pattern
Objectives	The subject develops students' ability to create sophisticated digital images that effectively represent fashion products with advanced techniques in digital rendering and image manipulation specific to fashion design. Students will be familiarised with the complete 3D design workflow, from concept to final visualisation, integrating 3D modelling into the broader fashion design process. Students will also be introduced to digital collaboration tools that facilitate teamwork in fashion design projects, with reference to the standard requirements and practices of the fashion industry.
Intended Learning Outcomes *(Note 1)*	Upon completion of the subject, students will be able to: a. Master advanced techniques in 3D sampling, fabric simulation, and virtual fitting. b. Create high-quality complex digital images that accurately represent fashion products and concepts; c. Be proficient in using digital collaboration tools to work effectively in team-based fashion design projects; d. Apply industry-standard practices in digital design and visualisation, ensuring their work meets professional expectations; and e. Critically analyse digital design challenges and propose effective solutions using advanced visualisation techniques.

Subject Synopsis/ Indicative Syllabus *(Note 2)*	**(I) Review of 3D Design Fundamentals** Overview of the basic 3D creation techniques and exploration of advanced features. **(II) Enhanced Fabric Simulation and Texturing** Techniques for simulating diverse fabric properties and behaviours, along with the application of custom textures and patterns to 3D garments. **(III) Precision Virtual Fitting** Conducting detailed virtual fittings with a focus on optimizing fit, and using advanced avatar customization for precise fitting assessments **(IV) Workflow Optimization** Streamlining the integration of 3D design into traditional fashion workflows, and acquiring suitable techniques for reducing lead times and minimizing waste through digital processes. **(V) Rendering and Presentation Mastery** Rendering techniques for high-quality visuals, including the exploration of lighting, shadows, and realistic environments **(VI) Collaboration and Review Platforms** Introduction to collaboration tools, with best practices for sharing and reviewing 3D samples online. **(VII) Industry Trends and Innovations** Analysing current trends in 3D design, including its limitation, virtual fitting, and technology in fashion. Exploring the potential of emerging technologies, such as AI and AR, in enhancing the design experience.

Teaching/Learning Methodology *(Note 3)*	Studio sessions will be used to deliver teaching in this subject with more interactive teaching and learning, and problem solving. Hands-on experience of CAD and demonstrations of digital design techniques will also be provided. Online learning materials and self-learning exercises with step-by-step instructions will also be incorporated via LEARN@PolyU to enhance students' understanding of apparel construction.

<table>
<tr><td rowspan="2">Assessment Methods in Alignment with Intended Learning Outcomes
(Note 4)</td><td>

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
1. In-class exercises	40%	✓	✓	✓	✓	✓
2. Projects	60%	✓	✓	✓	✓	✓
Total	100%					

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<tr><td>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Students will be expected to understand advanced digital 3D techniques in this subject. Such techniques will be acquired through extensive hands-on practice and exercises. Therefore, the course will be assessed by in-class exercises. Students will also be expected to communicate effectively and think critically by applying digital images and advanced 3D modelling techniques to support their future career, and such knowledge integration will be achieved through project work. In-class exercises and projects will be appropriate methods to assess students' proficiency in achieving the intended learning outcomes.</td></tr>
<tr><td>Student Study Effort Expected</td><td>

Class contact:	
▪ Studio	38 Hrs.
Other student study effort:	
▪ Self-study/Preparation	20 Hrs.
▪ Projects	50 Hrs.
Total student study effort	108 Hrs.

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Reading List and References	**Books**
	Ballery, C. & Conlon J. (2024). Fashion Business and Digital Transformation: Technology and Innovation across the Fashion Industry. Abingdon, Oxon: Routledge. Makryniotis, T. (2015). 3D Fashion Design: Technique, Design and Visualisation. London: Batsford. Matthews, J. L. (2011), Fitting and Pattern Alteration: A Multi-method Approach. Fairchild, New York. Pascal, V., & Nadia, M. T. (2000). Virtual Clothing. Theory and Practice. Sayem, A.S.M. (2023). Digital Fashion Innovations: Advances in Design, Simulation, and Industry. CRC Press. Song, G. (Ed.). (2011). Improving comfort in clothing. Elsevier. Von, N.R. & Tsang, J.H. (2023). Fashion Techn Applied Exploring Augmented Reality, Artificial Intelligence, Virtual Reality, NFTs, Body Scanning, 3D Digital Design, and More. Berkeley, CA: Apress: Imprint: Apress. **User Manual** https://support.clo3d.com/hc/en-us/categories/115000064648-Manual **Websites** CLO 3D, https://www.clo3d.com/ CLO-SET Connect, https://style.clo-set.com/aboutus

Note 1: Intended Learning Outcomes
Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus
The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology
This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.